सहयोगी प्राध्यापक (अन्न तंत्रज्ञान) व सहायक प्राध्यापक (अन्न तंत्रज्ञान व व्यवस्थापन)
महाराष्ट्र अभियांत्रिकी शिक्षक सेवा, गट-अ
उच्च व तंत्र शिक्षण विभाग
Associate Professor (Food Technology) and
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परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११२४)	इंग्रजी	१००	२००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक या विषयांमध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**FOOD CHEMISTRY**
	(a) Interactions amongst food constituents including those between the constituents themselves and with each other (water, protein, lipids, carbohydrates, minerals, vitamins), and the consequences thereof on nutritional, safety & sensory quality such as color & texture of foods. Examples include starch-lipid complexes, protein-polysaccharide interactions, protein-protein interactions leading to unnatural amino acids and racemization of amino acids, starch-polyphenol complexes. (b) Interactions of food constituents with external agents such as with flavours, food additives; among food additives; packaging materials and flavours; and the consequences thereof on nutritional, safety and sensory quality of foods (c) Contaminants generated during food processing (acrylamide, benzene, hydroxymethyl furfural, nitrosamines) and those that find their way in to foods as environmental contaminants (polychlorinated biphenyls, polychlorinated aromatic hydrocarbons, dioxins and furans, pesticide residues) (d) Anti-nutritional factors of significance in foods (trypsin inhibitors, phytates, tannins, hemagglutinins) and ways to mitigate them in products as legumes and millets (e) Microbial toxins of concern in foods and the effect of food processing therein. Examples are aflatoxin, patulin, bacterial toxins, zearalenone and such others. (f) Browning reactions in foods – Caramelization, enzymatic, non-enzymatic, ascorbic acid induced, and approaches to mitigate them. Impact of food processing on browning reactions, Role of browning reactions in foods; beneficial and adverse impacts of browning reactions on food organoleptic properties, browning and food quality, impact of browning reactions on nutritive value of foods. (g) Natural colors in foods: chemical structure, extraction, stability in food products and during processing (h) Flavours in foods – classification of flavours (natural, nature-identical and artificial); chemical pathways for generation of flavours in thermally processed and biochemical pathways for

	generation of flavours in fermented foods; quality assurance of flavours; selection of flavours for different types of foods, e.g. bakery, confectionary, microwaveable foods etc; off flavours and taints in foods
2	**FOOD BIOCHEMISTRY**
	(a) Carbohydrates: Fundamentals of chemistry of carbohydrates; Qualitative tests / colour reaction; Metabolic pathways and energy yield for breakdown of carbohydrates (b) Lipids: Types; Functions & comparative distribution of lipids; beta-oxidation; functions; quantitative analysis (c) Proteins: Chemistry; types, qualitative and quantitative analysis; Structural analysis (d) Nucleic acids: Types; Chemistry; Functions; Genetic code (e) Enzymes: definition, function, nomenclature, classification, mechanism of enzyme action, enzyme kinetics, enzyme inhibition and regulation (f) Vitamins & Co-enzymes: Structures& function of Nicotinamide, nicotinic acid, riboflavin, lipoic acid, biotin, thiamine, B6, folic acid, B12, pantothenic acid, ascorbic acid, vitamins A, D, K, and E
3	**FOOD MICROBIOLOGY**
	(a) Factors affecting spoilage of foods and associated microflora; Intrinsic and extrinsic factors affecting spoilage of foods; biochemical changes caused by microorganisms - putrefaction, lipolysis; Antagonism and synergism in microorganisms (b) Microbiological spoilage problems associated with typical food products such as dairy products, fruits and vegetables, grains and oilseeds, meat/fish and poultry, spices, and their control (c) Food borne infections and food poisoning, Microbial toxins, Emerging pathogens. (d) Detection methods for *E. coli*, *Staphylococci*, *Yersinia*, *Campylobacter*, *B. cereus*, *C. botulinum* &*Salmonella* from food samples. (e) Indicator organisms, microbiological quality assurance systems in food industry, use of the hazard analysis critical control points system to ensure microbiological safety and quality of foods, microbiological food standards (f) Rapid methods of microbial analysis; applications of immunological techniques to food industry
4	**FOOD NUTRITION**
	(a) Food composition and nutrients present in foods, terminologies used in nutrition, Food pyramid, my pyramid, my plate, Food exchanges and measures used, Energy value of foods, bomb calorimeter, physiological fuel value, estimation of energy value of foods from proximate composition. Basal Metabolic Energy, factors affecting, and calorie needs for B. M. E., physical activity and diet induced thermogenesis; energy imbalance and body weight regulation; Nutrition through lifecycle. (b) Role of carbohydrates in human nutrition- nutritionally important carbohydrates, physiological functions, digestion and absorption of available carbohydrates, dietary fiber, non-digestible oligosaccharides, resistant starch and its types, carbohydrates as prebiotics, dental carries and role of sugar, Lactose intolerance and galactosemia, Glycemic index and glycemic load of carbohydrate containing foods. (c) Role of lipids in human nutrition- nutritionally important lipid constituents, physiological functions, digestion and absorption of lipids, conjugated linoleic acid, trans fats, medium chain triglycerides, fat replacers and mimetics, keto diet, Cholesterol, Phytosterols, blood lipids (LDL, HDL, VLDL etc), essential fatty acids, their functions and deficiency, omega 3 and omega 6 PUFAs and their dietary sources, eicosanoids. (d) Role of proteins in human nutrition- essential/ non-essential amino acids, complete/ incomplete proteins, limiting amino acid, complementary proteins, physiological functions of proteins, daily protein requirements, digestion absorption and utilization of proteins, common food sources of proteins, protein deficiency (PEM/PCM)- prevalence, causes, effects, remedial measures (e) Concept and estimation of protein quality – *in vitro* (scoring methods, indices, microbiological methods, enzymatic methods) and *in vivo* methods (growth response methods like PER and Nitrogen balance methods like BV), PDCAAS (f) Anti-nutritional factors present in foods- antiproteins (trypsin inhibitor), antiminerals (phytate), anti- vitamins (ascorbic acid oxidase) and others- their chemistry, occurrence in food sources, mechanism of anti-nutritional action, processing stability, and remedial measures to reduce them. (g) Role of micronutrients (vitamins and minerals): ***Vitamins***– classification- water soluble (all the B vitamins and C) and fat soluble (Vitamins A, D, E and K); Chemistry, structure and

	properties; physiological functions; absorption and metabolism; food sources, deficiency and hypervitaminosis; RDA. ***Minerals-*** Nutritional and anti-nutritional mineral elements in food, their physiological role, food sources, deficiency diseases, bioavailability, Fortification of foods with vitamins and minerals. (h) Principles of Diet Therapy and Therapeutic Nutrition; Formulation of diets and foods for special needs; Techniques of diet and health surveys; Assessment of nutritional status; Effect of food processing, preservation and storage on nutritional quality of foods; Food nutrification; Sport nutrition;
5	**PRINCIPLES OF FOOD PRESERVATION**
	(a) Introduction to food preservation: Food spoilage; Underlying principles of different modes of food preservation; Preservation methods with emphasis on inactivation, inhibition, and avoiding recontamination. (b) Dehydration and drying of foods: Drying curve and drying time calculation; Water activity and moisture absorption isotherms; Psychometric chart; Different types of Dryers-Conductive, convective, and combined; IMF foods; osmotic dehydration. (c) Thermal processing of food products: Sterilization and Pasteurization; Canning of food products; Classifications and structure of cans, corrosion, Lacquering; Spoilage in canned foods (1) Thermal death time (TDT) concept; Process time calculation for canned foods; Retort processing; Aseptic packaging. (d) Low temperature storage and preservation: Chilling and Freezing; Freezing curve and water activity; Properties of frozen foods; Enthalpy change during freezing; Plank's equation for freezing time; Cold storage and Refrigeration load; Refrigeration cycle; Cryogenic freezing and IQF; Freeze concentration and freeze drying. (e) Hurdle technology: Role of acidity and pH in food preservation; Preservation by fermentation – Curing, Pickling and Smoking; Controlled and modified atmospheric packaging and storage; Chemical and Bio-preservatives; Antimicrobials.
6	**FOOD ANALYSIS**
	(a) Types of samples analyzed, steps in analysis, choice of methods; sampling procedures, considerations and sample preparation; Evaluation of analytical data – accuracy and precision, sources of errors, specificity, sensitivity and detection limits, regression analysis, reporting results. (b) Analysis of chemical constituents, their characterization and significance- moisture, ash, minerals, lipids, fat, proteins, fibre, titratable acidity, starch, reducing sugars. (c) Spectroscopic analysis of foods – basic principles, UV, visible, fluorescence, IR, AAS, MS, NMR. Chromatographic analysis of foods – basic principles, HPLC, GC, GLC, principles and applications. (d) Analysis of vitamins, extraneous matter, pesticides and mycotoxins. enzymatic, immunoassays, thermal analysis, and rheological profile.
7	**FOOD SAFETY AND REGULATIONS**
	(a) India Regulations Overview of FSSAI. Detail of FSS Regulations relevant to products and labelling. (FSS Licensing & Registration including Schedule IV, Product Standards & Additives, Labelling & Display, Advertisement & Claims) Introduction to food safety and security: Hygienic design of food plants and equipment. (b) Food quality: Various Quality attributes of food, Instrumental, chemical and microbial Quality control. Sensory evaluation of food and statistical analysis. Water quality and other utilities. (c) Food Safety Management System: Food Safety, Threat and Fraud Management based on international standards (HACCP / VACCP / TACCP)? (d) Global regulations: FAO, Technical Cooperation programmes, Bio-security in Food and Agriculture, World Health Organization (WHO), World Animal Health Organization (OIE), International Plant Protection, Convention (IPPC) Codex Alimentarius Commission - Codex India – Role of Codex Contact point, National Codex contact point (NCCP), National Codex Committee of India – ToR, Functions, Shadow Committees etc. (e) Overview of US and EU regulations for food
8	**FOOD ADDITIVES AND INGREDIENTS**
	(a) Additives in food processing and preservation, their functions and safety. (b) Safety and quality evaluation of additives and ingredients, acute and chronic studies, LD50. (c) Analytical methods, chemical and instrumental. (d) Various additives such as preservatives, antioxidants and sequestrants, colours and flavours and flavor enhancers, emulsifiers, humectants, hydrocolloids, stabilizers and sweeteners, acidulantsetc, with respect to chemistry, food uses and functions in formulations. New emerging additives, regulations as per CODEX and FSSAI. (e) Indirect food additives.

	(f) Colour additives in foods and their stability. (g) Classification of flavours and the process of preparing including extraction, distillation, fractionation and purification; Stability of flavours. (h) Ingredients used in food production e. g. sugars, starches, proteins/protein hydrolysates/isolates, fats, prebiotic oligosaccharides, pectin, chitin, and their technology of production and application, unusual protein sources such as insect proteins, mycoproteins
9	**FOOD PACKAGING**
	(a) Introduction to food packaging; Causes of food spoilage; Factors affecting food spoilage; Packaging as a method for preservation of foods; Functions of food packaging including new packaging technologies; Levels of packaging; Food labelling. (b) Different materials used in food packaging such as paper, board, glass, metal containers, aluminium foil, plastics, composites, traditional materials and their physicochemical characteristics, their advantages and limitations, method of manufacture. (c) Testing of various packaging materials and packages for evaluation of quality, for identification, for evaluation of barrier and strength properties for transport-worthiness, for biodegradability, for migration etc; Criteria for selection of packaging materials; Shelf-life testing of packaged foods. (d) Food and packaging material interactions including migration, scalping of flavour; biodegradable packaging; application of nanotechnology in food packaging; environmental concerns and lifecycle assessment. (e) Packaging of various food commodities including fresh produce (fruits and vegetables), meat, fish, poultry, milk, cereals and processed foods. (f) Newer packaging technologies- CAP/MAP packaging; aseptic processing and packaging; irradiated packaging; retort pouch; microwaveable packaging; packaging for non-thermal processing of foods (pulsed electric field, high pressure processing, irradiation etc.); active packaging; intelligent packaging. (g) Emerging trends in polymers for food packaging, green packaging materials, nanoparticles in food packaging. (h) Recycling of food packaging materials and use of recycled materials in food packaging, edible packaging.
10	**TECHNOLOGY OF FRUITS AND VEGETABLES**
	(a) Fruits and Vegetables: Types, Structure and composition, development, maturity indices, Methods, importance and overall quality of fruit and vegetables for harvesting. (b) Post-harvest Processing: Handlig, storage, ripening and control of ripening, chemical changes etc. of fruits and vegetables (c) Fruits: Processing techniques, juices, juice extraction process, causes of juice spoilage, Methods of juice preservation, concentrates, preserves, Jams/Marmalades, Squashes/cordials, Candied Fruits Fruit Bar, and other traditional products. (d) Vegetables: Processing techniques, vegetable juices, preservation, Ketchup/sauces, Chutneys, Soup powders, pickles, fermented pickles and other traditional products. Dried powders (Onion, garlic, potato, carrot starch), dried Cauliflower and cabbage: Sauerkraut, Pickles, Dried Leafy Vegetables. (Spinach, Fenugreek, Coriander leaves, Curry leaves). Bitter gourd: Pickle, Dried bitter gourd. (e) Tubers: Processing and products (Potato, Sweet potato, turnips, beetroot, Taro, yam and others) (f) Dehydrated and specialty products and by-products of fruits and vegetables (g) Honey, Sugars and saccharine products. Soft drinks, fermented pickles.
11	**TECHNOLOGY OF TEA, COFFEE AND COCOA**
	(a) Cultivation of cocoa, microbiology/biochemistry of cocoa bean fermentation, development of cocoa flavour precursors, drying, roasting, alkalization (Dutching), NARS process, winnowing, nib grinding, liquor processing, expeller pressing, cocoa cake grinding, cocoa products, analysis of cocoa powder, cocoa butter-chemistry and properties. (b) Cocoa butter replacement fats (CBS and CBE fats and other fats), antibloom fat, lecithin as emulsifier in chocolate; Chocolate based Confectionery-Bulk chocolate manufacture: raw materials, milk chocolate process including milk crumb process, melangeuring, refining, pasting, conching; tempering, moulding, enrobing, panning. (c) Tea cultivation, constituents of tea leaf, chemistry and technology of fermentation and black tea manufacture, characteristics and quality of tea beverage, types of tea- black tea, green tea, oolong tea, speciality teas, herbal teas; analysis of tea, iced tea, other tea-based products, kombucha tea. (d) Coffee varieties, cultivation, coffee bean composition, processing of berries (wet and dry process), roasting, grinding, brewing, instant coffee manufacture, decaffeination; analysis of coffee and chicory, green coffee, filter coffee and its packaging, health benefits of coffee

12	**TECHNOLOGY OF CEREALS, LEGUMES AND OILSEEDS**
	(a) Cereals and millets: Morphology: physicochemical properties; chemical composition and nutritional value. (b) Wheat: Milling (chakki, roller) break system, purification system and reduction system; Maida, suji, chakki atta, extraction rate and its effect on flour composition; damaged starch, Quality characteristics of flour and their suitability for baking variety of products. (c) Rice: Paddy processing and rice milling: conventional milling, modern milling operations, milling machines, milling efficiency, byproducts of rice milling. Quality characteristics influencing final milled products. (d) Parboiling: Parboiling of rice, Rice bran stabilization and its methods; Aging of rice; Enrichment – need, methods; processed foods from rice – breakfast cereals, flakes, puffing, canning and instant rice (e) Corn: Corn milling – dry and wet milling, starch and gluten separation, milling fractions and modified starches, corn grits and flakes (f) Barley: Pearling / milling, Malting process, malt based foods. (g) Oats: Processing and oats milling, Flaked oats in breakfast cereals (h) Sorghum: Milling, Malting, Pearling and industrial utilization (i) Millets: Importance of Millet, composition, processing of millets for food uses, major and minor millets (j) Bakery Products: Breads, Cakes, Biscuits, Different types of biscuits, short, hard and fermented Biscuits, Advanced bakery products such as croissants, puffs, muffins and filled cookies and muffins, Different functional ingredients used in baking, Gluten free, multigrain products (k) Processing and technology of legumes and oilseeds: Moong, Channa, Arhar, Urd, whole as well split dal. Technology of oilseeds such as peanut, sesame, sunflower etc, Utilisation in food industry as protein and oil source and their use in Indian diet.
13	**TECHNOLOGY OF MILK AND MILK BASED PRODUCTS**
	(a) Milk components, composition and types; raw milk quality and processing (HTST/UHT, homogenization), CIP/COP/distribution/packaging/fermented milks/bulk starter cultures (b) Manufacture of milk-based products - condensed and evaporated milk, milk powder, cheese, ice-cream, cream, butter, ghee; their evaluation and quality parameters, defects encountered during production, packaging and storage. (c) Non-dairy milk and milk products; Casein and caseinates, lactose, whey protein concentrates and isolates, milk co-precipitates, and other specialty products (d) Traditional dairy products, milk confections such as yoghurt, *dahi, khoa, burfi, kalakand, gulabjamun, rosogolla, shrikhand, chhana, paneer, ghee, lassi*etc. (e) Probiotic milk products.
14	**TECHNOLOGY OF PLANTATION PRODUCTS**
	(a) Varieties of spices/condiments grown and consumed in various countries including India, nomenclature, properties and culinary uses, preservative action, medicinal uses, analysis of spices, post-harvest handling/ storage/ preservation/ processing of spices, spice-based products, major individual spices- turmeric, cardamom, asafoetida, cinnamon, cloves, nutmeg, capsicum, pepper, ginger, saffron, anise, ajwain, coriander, cumin, celery, caraway, dill, fenugreek, fennel etc. (b) Sugar based confectionery- ingredients used including sugar and alternative sweeteners, description of types of confections like HBC, toffee, fudge, gums and jellies, aerated confectionery, sugar panned confections, chewing gum etc, manufacturing process and equipment, structure of sugar confection, chemical analysis and quality assurance. Indian confectionery- types, description of characteristics, method of preparation (c) Cashewnut chemistry, technology, processing and by-products – cashewnut shell liquid, nuts, cashew apple (d) Coconut processing in to value added products- coconut water, coconut milk, dessicated coconut, coconut oil, grated coconut and their products; by-product utilization; Other lesser-known nuts such as arecanut, rubber, cinchino and palmyrah (e) Commercial floriculture - Production and trade of cut flowers, foliage plants, potted plants, landscape plants, bedding plants, seed production, dried flowers and plant parts, perfumes and essential oils and natural dyes for viable agri-business option, extension of shelf life of flowers, edible flowers and their bioactivities. (f) Medicinal plants of commercial importance in nutraceutical formulations, processing, formulations and applications, quality assurance
15	**TECHNOLOGY OF ANIMAL BASED PRODUCTS**
	(a) Slaughter of food animals: Plan and layout of slaughterhouse, Religious and scientific methods of Slaughter of Food animals; Principle and Methods of Stunning of Food Animals

	(b) Handling and Transport of Food Animals: Animal welfare and pre-slaughter care, handling and transport of meat animals including poultry. Stress and Meat quality (DFD and PSE conditions) (c) AM and PM and Dressing of Food Animals:Procedures of Ante-mortem and post mortem examination of meat animals, Emergency and casualty slaughter. (d) Structure, Composition and Nutritive Value of Meat, Postmortem changes in meat: Conversion of muscle to meat, Ageing of meat (e) Meat Quality: Factors affecting microbial growth, spoilage of meat; Physicochemical parameters of meat, colour, texture, Sensory evaluation of meat and meat products (f) Meat Processing, preservation and packaging: Meat processing- Basic processing techniques of meat viz. Smoking and Curing, emulsification; Preparation of meat products including fermented meats, Preservation of meat and meat products; Packaging of meat and meat products. (g) Meat plant hygiene: GMP and HACCP. National and International Laws related to meat (h) Slaughterhouse by products: By-products from meat industries and their utilization (i) Meat trade: Statistics of meat industry in India (j) Poultry meat and Egg: Composition and nutritional value of poultry meat and eggs, Processing of poultry meat and eggs; Preparation of different poultry and egg products, Evaluation of external and internal quality of egg; Egg preservation, Utilization of poultry by products, Microbial Spoilage and control. (k) Fish and Fish Products: Classification of freshwater fish and marine fish; Commercial handling, storage and transport of raw fish. Average composition of fish; Freshness criteria and quality assessment of fish; Spoilage of fish. Methods of processing and preservation of fish-Canning, Freezing, Drying, Smoking and Curing. Fish products – fish meal, fish protein concentrate, fish liver oil, fish sauce and surimi; Fish processing industries in India. (l) Cultured meats and plant-based alternatives
16	**FOOD QUALITY MANAGEMENT**
	(a) Food Quality concept, Service quality; quality planning, control and improvement; ISO 9000 standards; quality and financial performance, quality failure; (b) Design of quality control system, Process quality control, attribute control, variables control, control charts; continuous improvement, six sigma, lean and six sigma, quality control and improvement in food industry.
17	**FOOD SUPPLY CHAIN MANAGEMENT**
	(a) Introduction and overview of supply chain management: Agriculture and food supply chain management, supply chain as a source of competitive advantage, performance measurement in agri-food supply chains. (b) Management challenges in cold chain: Post-harvest food management – supply chain management, the major cold chain technologies dry ice, gel packs, eutectic plates, liquid nitrogen, quilts, reefers refrigerated containers, managerial economics – fresh food and supply chain management challenges, life cycle assessment studies of food product, retail logistics changes and challenges retail logistics and supply chain transformation the cold chain and its logistics, from a geographical perspective, the cold chain has the following impacts the major cold chain technologies, value chain, global value chain. (c) Risk and Uncertainty: Types and sources of risk in agricultural & food supply chain, managing risk and uncertainty, FMEA, method for coping with bullwhip effect, supply chain integration, push vs. pull systems. Managing Inventories and Coordination: Inventory management, types of inventories, inventory management, economic order quantity (EOQ), inventory classification and control – ABC Analysis. (d) Logistics & Strategic Alliances: Inbound and outbound logistics, 3rd/4th Party Logistics (3PL/4PL), cross docking, retailer-supplier partnerships, buyer vendor coordination, procurement, vendor development and evaluation-factor rating method. Outbound logistics: Designing supply chain network, system view of logistics-coordination and management of transportation, inter model transportation and third-party transportation services, characteristics of different transportation services, distribution (e) strategies. (f) Procurement & Outsourcing Strategies: Buy-make decision, procurement strategy, framework of e-Procurement. (g) Strategic considerations for supply chain, Porter's industry analysis and value-chain models, Reverse Logistics.
18	**FOOD HYGIENE AND SANITIZATION MANAGEMENT**
	(a) Basics of food hygiene, Personal hygiene, Food handling habits, handling potable water supply. Microbial standards for foods, food poisoning, food-borne illness, food allergy, contamination,

	hazard and HACCP. (b) Food safety issues and strategies, Factors affecting food safety, Importance of safe foods, factors affecting shelf life and methods to check the shelf life. (c) National Food Legislations: PFA, FPO, MPO, BIS, AGMARK, ISI, Misbranding, Factors contributing to physical, chemical and biological contamination in food chain, prevention and control of food borne hazards, Regulation of food sanitation, sources of contamination, personal hygiene of food handlers, Method of cleaning and disinfection. (d) Detergents and Sanitizer. Sanitation in food service. cleaning compounds, sanitation methods, waste disposal strategy (solid and liquid waste) and pest control, Good Manufacturing Practices (GMP) and Good Hygiene Practices (GHP) by food businesses General hygienic and sanitary practices Petty food operators and Street food vendors, VACCP (Vulnerability Assessment Critical Control Point). TACCP (Threat Assessment Critical Control Point).
19	**NUTRACEUTICALS AND FUNCTIONAL FOODS**
	(a) Introduction to nutraceuticals: definitions, synonymous terms, basis of claims for a compound as a nutraceutical, regulatory issues for nutraceuticals including CODEX; nutrigenomics - an introduction and its relation to nutraceuticals (b) Clinical testing of nutraceuticals and functional foods; interactions of prescription drugs and nutraceuticals; adverse effects and toxicity/safety of nutraceuticals (c) Nutraceuticals/ functional foods for life-style associated diseases such as atherosclerosis, hypertension, heart disease, stroke, obesity, type 2 diabetes, and diseases associated with smoking, alcohol and drug abuse and their mechanisms of action, dosage levels, contraindications if any. (d) Nutraceuticals/ functional foods for aging associated diseases such as cancer, arthritis, cataract, osteoporosis, Alzheimer's disease, age related macular degeneration and their mechanisms of action, dosage levels, contraindications if any. (e) Manufacturing aspects of selected nutraceuticals such as lycopene, isoflavonoids, prebiotics and probiotics, glucosamine, phytosterols etc.; formulation of functional foods containing nutraceuticals, stability, and analytical issues, labelling issues
20	**NEWER TECHNIQUES IN FOOD PROCESSING AND PRESERVATION**
	(a) Newer techniques in thermal processing: Concept of HTST; UHT; Ohmic, Dielectric, Infra-red Heating; Microwave heating; Frying method. (b) Non-thermal processing of food: High pressure processing; Pulsed electric field processing; Cold extrusion; Plasma processing; Ionizing Radiations; Ultrasound processing; UV and Pulsed light processing; Membrane Technology.
21	**WASTE MANAGEMENT IN FOOD INDUSTRY**
	(a) Water quality, treatment and recycle. BOD, COD and definitions, Discharge limits for effluents. Primary treatment, secondary and tertiary treatments by physical, chemical and biological methods. (b) Effluent and solid waste utilization food processing industry by biological methods – for SCP, biogas and other products (c) Waste management strategies and value-added products from of agri-food processing industry (d) Recovery of biological from dairy, meat, fish and poultry processing industry (e) Case studies: Cane sugar waste, molasses for alcohol, bagasse for paper pulp, chemicals, bioethanol, cogeneration. Other processes including vermiculture.

दिनांक – ११ फेब्रुवारी, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग